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Start of meeting

1 Progress and Pre-Meeting Notes

The high-resolution (10Hz) spindle torque datasets have been successfully acquired, unlocking new modeling perspectives.

- **Preprocessing Phase:** The data cleansing pipeline must be refactored. All plots, charts, and data visualizations need to be regenerated using this new 10Hz dataset.
- **Modeling Objective:** Upcoming tests and benchmarks will start to be focused on the TFT (Temporal Fusion Transformer) model.
- **Thesis Narrative Strategy:** Regarding the exploratory data analysis section, a **narrative and iterative approach** was approved. Instead of hiding the old data steps, the manuscript will chronologically describe the research journey: initial data, early analysis (Spearman correlation, PCA), identification of limitations, and the eventual pivot to the new 10Hz high-resolution dataset.

2 Feature Engineering and Data Processing

2.1 Filtering and Feature Selection

- **Roughing Phase Management:** It is crucial to evaluate the impact of different machining steps. Comparative tests must be conducted by including and excluding the **Roughing** phases, strictly focusing evaluation on **Pre-Finishing** and **Finishing** operations.
- **Blanking Management:** Special care must be taken to handle and filter out blanking data (air-cutting/ idle phases) to eliminate non-relevant noise.
- **Pass Type Feature Relevance:** Including the raw "Pass Type" column as a model input is not highly critical, as its information is already redundant with other numeric features in the dataset.

2.2 Prediction Horizon

- **Lookback Parameter:** Further reflection and empirical validation are required to define the optimal historical window length (evaluating sequences of 5 to 10 past passes).

3 Benchmarking and Optimization Framework

- 25 A strict, empirical, and quantifiable comparison protocol must be formalized for the thesis report.
- **Hyperparameter Tuning:** Optimization must be systematically performed for every model. The use of **Optuna** integrated with the **Darts** library is approved to discover the best training hyperparameter configurations using validation metrics.
 - **Leveraging the “Known Future” (TFT):** Feature selection must be rigorously justified.
30 The model will exploit the planned operational context of the machine, since at step t comes back: (3 / 3) present 1, almost all parameters are deterministically known (pass type, alloy, spindle speed, depth of cut), with the sole exception of the spindle torque (y).
 - **Acoustic Data Ablation Study:** To scientifically measure the impact of audio data, the benchmarking system must systematically evaluate each model architecture under two distinct configurations: **with**
35 and **without** sound features. The main objective is to establish an empirical and quantifiable proof regarding the value of the acoustic modality for the final report.

11:30 PM Meeting Adjourned

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